

Teacher's Answer Key

Chapter 16: Capacitors

1. A (Armatures separated by a dielectric)

Rationale: In physics, the conducting plates of a capacitor are specifically termed "armatures", and the insulating medium that sits between them to prevent current flow and increase capacity is called the "dielectric".

2. C ($60 \mu C$)

Rationale: The stored charge is the direct product of the capacitance and the applied voltage, calculated as $Q = 5 \times 10^{-6} F \times 12 V = 60 \times 10^{-6} C$.

3. C ($2 mJ$)

Rationale: Applying the energy formula $W = \frac{1}{2}CU^2$ yields $0.5 \times 10 \times 10^{-6} \times (20)^2 = 0.5 \times 10^{-5} \times 400 = 0.002 J$, which is $2 mJ$.

4. C ($44.25 pF$)

Rationale: Using the geometric capacitance formula $C_0 = \epsilon_0 \frac{S}{d}$, where $S = (0.1 m)^2 = 0.01 m^2$ and $d = 0.002 m$. Thus, $C_0 = 8.85 \times 10^{-12} \times \frac{0.01}{0.002} = 44.25 \times 10^{-12} F = 44.25 pF$.

5. D ($5.4C_0$)

Rationale: Introducing a dielectric increases the original vacuum/air capacitance by a multiplying factor equal to the material's relative permittivity (ϵ_r).

6. A ($3 \mu F$)

Rationale: The inverse of the equivalent series capacitance is the sum of the inverses: $1/C_{eq} = 1/4 + 1/12 = 3/12 + 1/12 = 4/12$. Taking the reciprocal gives $C_{eq} = 12/4 = 3 \mu F$.

7. B (A $5 \mu F$ capacitor)

Rationale: In a parallel configuration, individual capacitances simply add together to form the total equivalent capacitance ($C_{eq} = C_1 + C_2$). Thus, $15 = 10 + C_2$, making $C_2 = 5 \mu F$.

8. B ($1.0 s$)

Rationale: The time constant is the product of resistance and capacitance. Ensuring conversion to standard SI units: $\tau = 20,000 \Omega \times 0.000050 F = 1 s$.

9. C (6.3 V)

Rationale: During the charging phase, the capacitor reaches approximately 63% of the source EMF ($10\text{ V} \times 0.63 = 6.3\text{ V}$) after exactly one time constant has elapsed.

10. B (8.9 V)

Rationale: During discharge, the voltage drops exponentially, leaving approximately 37% of its initial maximum value after one time constant passes ($0.37 \times 24\text{ V} \approx 8.88\text{ V}$).

11. C (It doubles.)

Rationale: Capacitance is inversely proportional to the distance between the plates ($C \propto 1/d$). Dividing the denominator by 2 mathematically multiplies the overall value by 2.

12. D (4W)

Rationale: Because stored energy is proportional to the square of the voltage ($W = \frac{1}{2}CU^2$), doubling the voltage increases the energy by a factor of $2^2 = 4$.

13. C (6.0 μF)

Rationale: First, find the equivalent of the series branch (C_1 and C_2): $C_{12} = \frac{6 \times 3}{6+3} = 2\text{ }\mu\text{F}$. Then, add this in parallel with the third capacitor (C_3): $C_{eq} = C_{12} + C_3 = 2 + 4 = 6\text{ }\mu\text{F}$.

14. A ($Q_1 = 24\text{ }\mu\text{C}$, $U_1 = 4\text{ V}$, $W_1 = 48\text{ }\mu\text{J}$)

Rationale: The series branch (C_1, C_2) has 12 V across it and an equivalent capacitance $C_{12} = 2\text{ }\mu\text{F}$. The total charge on this branch is $Q_{12} = C_{12} \times U = 2\text{ }\mu\text{F} \times 12\text{ V} = 24\text{ }\mu\text{C}$. Because they are in series, $Q_1 = Q_2 = Q_{12} = 24\text{ }\mu\text{C}$. The voltage across C_1 is $U_1 = Q_1/C_1 = 24\text{ }\mu\text{C}/6\text{ }\mu\text{F} = 4\text{ V}$. Its energy is $W_1 = \frac{1}{2}C_1U_1^2 = 0.5 \times 6\text{ }\mu\text{F} \times (4\text{ V})^2 = 48\text{ }\mu\text{J}$.